

MongoDB Compass - CSIT/Shell

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

CSIT

- admin
- bankdb
- config
- local
 - startup_log

localhost:27017

mongosh: CSIT

>_MONGOSH

```
title: 'quantum horizons',
author: 'R. Menon',
genre: 'Science',
publishedYear: 2023,
formats: [
  'Digital',
  'Audiobook'
],
specs: {
  fileSize_MB: 5.6,
  drm: true,
  narrator: 'P. Iyer',
  runtime_min: 540
},
avgRating: 4.9,
reviews: [
  {
    member_id: 'M4001',
    rating: 5,
    comment: 'The Digital format made annotations so easy.',
    date: 2023-08-01T00:00:00.000Z
  },
  {
    member_id: 'M4002',
    rating: 5,
    comment: 'Mind-bending and well narrated.',
    date: 2023-09-14T00:00:00.000Z
  }
]
}
bookflow_db>
```

```

db.book_metadata.find(
  {'formats':{'regex':"Digital",$options:"i"}},
  {'title:1,_id:0}
).forEach(x=>print(x.title));

print("Q5: Digital in Review");
db.book_metadata.find(
  {"reviews.comment":{"regex":"Digital",$options:"i"}},
  {'title:1,_id:0}
).forEach(x=>print(x.title));
Q1: Rating > 4
The Silent Algorithm | 4.6
Ocean of Code | 4.2
Quantum Horizons | 4.9
Q2: Digital or Audiobook
The Silent Algorithm
Ocean of Code
Quantum Horizons
Q3: Rating > 4 AND Digital
The Silent Algorithm
Ocean of Code
Quantum Horizons
Q4: Digital using Regex
The Silent Algorithm
Ocean of Code
Quantum Horizons
Q5: Digital in Review
The Silent Algorithm
Ocean of Code
Quantum Horizons
lookflow db>

```

```

> print("SS3: Top Rated Books");

db.book_metadata.aggregate([
  {$match:{publishedYear:{$gt:2020}}},
  {$unwind:"$reviews"},
  {$group:{
    _id:"$title",
    author:{$first:"$author"},
    avgReviewRating:{$avg:"$reviews.rating"},
    totalReviews:{$sum:1}
  }},
  {$sort:{avgReviewRating:-1}},
  {$project:{
    _id:0,
    title:"$_id",
    author:1,
    avgReviewRating:{$round:["$avgReviewRating",2]},
    totalReviews:1
  }}
]).forEach(x=>print(
  x.title+" | "+x.author+" | Avg Rating: "+x.avgReviewRating+" | Reviews: "+x.total
));

```

```

< SS3: Top Rated Books
< Quantum Horizons | R. Menon | Avg Rating: 5 | Reviews: 2
< Ocean of Code | A. Fernandes | Avg Rating: 4.5 | Reviews: 2
< The Silent Algorithm | R. Menon | Avg Rating: 4.5 | Reviews: 2

```

```
> print("SS4: Text Index Search");

db.book_metadata.createIndex({"reviews.comment":"text"});

db.book_metadata.find(
  {$text:{$search:"Digital"}},
  {title:1,_id:0}
).forEach(x=>print(x.title));
< SS4: Text Index Search
< The Silent Algorithm
< Quantum Horizons
< Ocean of Code
bookflow_db> |
```

```
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'backward',
        indexBounds: {}
    }
}
},
rejectedPlans: []
},
executionStats: {
    executionSuccess: true,
    nReturned: 3,
    executionTimeMillis: 12,
    totalKeysExamined: 3,
    totalDocsExamined: 3,
    executionStages: {
        isCached: false,
        stage: 'TEXT_MATCH',
        nReturned: 3,
        executionTimeMillisEstimate: 0,
        works: 4,
        advanced: 3,
        needTime: 0,
        needYield: 0,
        saveState: 0,
        restoreState: 0,
        isEOF: 1,
```

```
executionStats: {
  executionSuccess: true,
  nReturned: 3,
  executionTimeMillis: 9,
  totalKeysExamined: 3,
  totalDocsExamined: 3,
  executionStages: {
    isCached: false,
    stage: 'FETCH',
    nReturned: 3,
    executionTimeMillisEstimate: 1,
    works: 4,
    advanced: 3,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'bookflow_db.book_metadata',
    docsExamined: 3,
    alreadyHasObj: 0,
    inputStage: {
      stage: 'IXSCAN',
      nReturned: 3,
      executionTimeMillisEstimate: 1,
      works: 4,
      advanced: 3,
      needTime: 0,
      needYield: 0,
      saveState: 0,
```